RAJALAKSHMI ENGINEERING COLLEGE

Abirami G 251001004-2025-IT

A2

Started on	Tuesday, 25 August 2026, 11:18 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:26 AM
Time taken	7 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n, i, s, count = 0;

    scanf("%d", &n);

    i = 1;
    count++;

    s = 1;
    count++;

    while (s <= n)
    {
        count++;

        i++;
        count++;

        s += i;
        count++;
    }

    count++;   // final while condition

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course